

The double-stranded lattice sum as a fermionic partition function

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Abstract

Bio 12 counted the lattice points of the double-stranded box of ϕ X174 at $k = 10$, and those with connected support, but not the number $\#DS$ of double-stranded reconstructions. The obstacle is the arborescence factor $t(G_c)$, a determinant of the whole support. We show that it is a fermionic partition function with one *even* factor per arc, so that it can be accumulated arc by arc in any order. A reduced determinant needs a root, and the support moves with the point. Both difficulties are removed together by the reverse-complement symmetry $c \mapsto m - c$, which fixes the summand and halves the box along any orbit of odd multiplicity, where one fixed vertex is always in the support. The weight is made integral by trading $\prod_a c(a)!^{-1}$ for binomials and one constant. Everything is checked exactly against Bio 8 at $k = 11, 12$ and on random covers. Frontier elimination of the fermionic weight is exact but does not reach $k = 10$, because its state is linear and a single history already fills up to 3^w of its coordinates. Nor does the positive (arborescence) form of the same weight: it has one state per history, but merges too few of them. The number is obtained instead by visiting all 1 240 843 950 points of the half-box in compiled code, contracting forced arcs so that only a small determinant is left at each point, and working modulo five primes. The run reproduces both of Bio 12's counts, halved, and agrees with the exact arborescence elimination on $k = 10$ sub-boxes. The result is

$$\#DS(\phi X174, 10) = 31\,925\,753\,246\,212,$$

the last blank entry of Bio 8's Table 2.

Keywords. de Bruijn graph, reverse-complement double cover, circulation lattice, matrix-tree theorem, Berezin integral, Grassmann variables, transfer matrix, frontier elimination, Eulerian circuits, multimodular arithmetic, genome assembly.

1 What is left to count

We keep the notation of [1] and [2]. S is a circular sequence, D_k the labelled reverse-complement double cover, A its set of arc contents, $m(a)$ the number of labelled arcs of content a , and ρ the reverse-complement involution, so that $m(\rho a) = m(a)$ and D_k is balanced. A circular sequence T is a *double-stranded reconstruction* of S at word length k when $\text{spec}(T) + \text{spec}(\bar{T}) = \text{spec}(S) + \text{spec}(\bar{S})$; writing $c = \text{spec}(T)$ this says that c is a circulation with $c + \rho c = m$, and with $f = 2c - m$ that

f lies in the (-1) -eigenlattice $\Lambda^- = \{f \in Z_1(D_k) : \rho f = -f\}$, inside the box $|f| \leq m$ and in the parity class of m . Hence [1, Thm. 7]

$$\#DS(S, k) = \sum_f N_{\text{seq}}\left(\frac{m+f}{2}\right), \quad N_{\text{seq}}(c) = \frac{t(G_c) \text{Loc}(G_c)}{\prod_a c(a)!}, \quad (1)$$

where G_c is the multidigraph with $c(a)$ copies of each content, $t(G_c)$ its number of spanning arborescences to a root, and $\text{Loc}(G_c) = \prod_v (d_c^+(v) - 1)!$.

Bio 12 observed that the constraint defining the sum is one linear equation per vertex, each involving only the four endpoints of one ρ -orbit, and replaced Bio 8's backtracking by a *frontier elimination*: process the orbits in a good order carrying, as state, the partial imbalance at the vertices still open. At $k = 10$, where $\text{rank } \Lambda^- = 48$, that counts the 2 481 687 900 points of the box in seconds with a frontier 26 wide and 50 758 states, and, carrying in addition the partition of the frontier into the classes already joined, the 909 103 210 points whose support is connected.

It did not reach $\#DS$. The reason is entirely in $t(G_c)$: the number of spanning arborescences is the determinant of a reduced Laplacian of the whole support — of order 125 at $k = 10$ — and a determinant is not a product of local scalars. A frontier state that is a number, or a partition, cannot carry it.

The point of this note is that it *is* a product of local *fermions*.

2 The determinant as a product over arcs

Let V be the vertex set, $r \in V$ a distinguished *root*, and let $\Lambda = \Lambda(\bar{\psi}_v, \psi_v : v \in V \setminus \{r\})$ be the Grassmann algebra on two odd generators per non-root vertex; we put $\psi_r = \bar{\psi}_r = 0$. For $c \in \mathbb{Z}_{\geq 0}^A$ let L_c be the out-Laplacian of G_c , $L_c(v, v) = d_c^+(v) - (\text{loops at } v)$ and $L_c(v, u) = -\sum_{a: t(a)=v, h(a)=u} c(a)$ for $u \neq v$, and let

$$A_c = L_c + \text{diag}(\mathbf{1}_{v \notin \text{supp } c}), \quad \text{supp } c = \{v : d_c^+(v) > 0\}, \quad (2)$$

so that A_c agrees with L_c on $\text{supp } c$ and is the identity off it. Write $\tilde{A}_c^r$ for A_c with the row and the column of r deleted.

Proposition 1 (arc factorisation). *In Λ ,*

$$P_c = \prod_{a \in A} \left(1 - c(a) \bar{\psi}_{t(a)}(\psi_{t(a)} - \psi_{h(a)})\right) \cdot \prod_{v \notin \text{supp } c} (1 - \bar{\psi}_v \psi_v) = (-1)^{|V|-1} \det \tilde{A}_c^r \cdot \Psi, \quad (3)$$

where $\Psi = \bar{\psi}_{v_1} \psi_{v_1} \bar{\psi}_{v_2} \psi_{v_2} \cdots$ runs over $V \setminus \{r\}$ in any order. Every factor in (3) has even degree, so the product may be formed in any order whatsoever, one arc at a time.

Proof. Each factor is $1 + x$ with x a linear combination of $\bar{\psi}_v \psi_u$, so $x^2 = 0$ and $1 + x = e^x$; and $\bar{\psi}_v^2 = 0$ makes the product of the factors belonging to one tail v equal to $1 - \bar{\psi}_v \sum_u A_c(v, u) \psi_u$, the loops cancelling between the two terms of $\psi_{t(a)} - \psi_{h(a)}$. Hence $P_c = \prod_{v \neq r} (1 - \bar{\psi}_v (A_c \psi)_v) = \exp(-\sum_{v, u} \bar{\psi}_v A_c(v, u) \psi_u)$, the Berezin integrand of $\det \tilde{A}_c^r$. Expanding, $P_c = \sum_{\sigma} \prod_v (-A_c(v, \sigma v)) \bar{\psi}_{v_1} \psi_{\sigma v_1} \bar{\psi}_{v_2} \psi_{\sigma v_2} \cdots$, the sum over the bijections σ of $V \setminus \{r\}$. Each block $\bar{\psi}_v \psi_{\sigma v}$ is even, so the blocks commute and the vertex order chosen for Ψ is immaterial; and interchanging two of the ψ 's moves an odd generator across a block of odd length, which costs -1 , so the monomial equals $\text{sgn}(\sigma) \Psi$. The $(-1)^{|V|-1}$ is the one minus sign per row. $\square$

Remark 2. The constant $(-1)^{|V|-1}$ in (3) is attached to the *interleaved* order $\bar{\psi}_{v_1}\psi_{v_1}\bar{\psi}_{v_2}\psi_{v_2}\cdots$. Two interleaved orders differ by a permutation of the even blocks $\bar{\psi}_v\psi_v$ and so give the same constant, but listing all the $\bar{\psi}$ first and all the ψ afterwards does not: that costs a shuffle sign which, when the two lists are ordered differently — as they are in an implementation where $\bar{\psi}_v$ and ψ_v have different lifetimes — varies from instance to instance. This is not a scruple; it is the one place where an implementation of (3) silently returns $-\#DS$.

The point of Proposition 1 for the frontier is this. The factor of the arc a involves only $\bar{\psi}_{t(a)}$, $\psi_{t(a)}$ and $\psi_{h(a)}$, and it is even. So the whole determinant can be accumulated in Bio 12's orbit order, one ρ -orbit at a time, and what has to be carried across the frontier is: for each open vertex v , whether $\bar{\psi}_v$ has already been spent and whether ψ_v has already been taken. Two bits per vertex, not a set partition. Section 6 explains why this turns out to be the more expensive of the two alphabets.

3 The root, and why the box is halved

A reduced determinant needs a root, and $\text{supp } c$ moves with c : no vertex of D_k lies in every support. The diagonal completion (2) takes care of the vertices that are missing, and the following lemma says what $\tilde{A}_c^r$ then computes.

Lemma 3. *Let c be a circulation with $c \neq 0$. If $r \in \text{supp } c$ then $\det \tilde{A}_c^r = t(G_c)$, independently of which $r \in \text{supp } c$ is taken; and it vanishes exactly when $\text{supp } c$ is disconnected. If $r \notin \text{supp } c$ then $\det \tilde{A}_c^r = 0$.*

Proof. c is a circulation, so $d_c^- = d_c^+$ and no arc of G_c leaves $\text{supp } c$ or enters it: A_c is block diagonal, L_c on $\text{supp } c$ and the identity off it. Deleting $r \in \text{supp } c$ leaves $\det$ equal to the corresponding reduced Laplacian of G_c , which is $t(G_c)$ by the matrix-tree theorem; G_c is balanced, so its weak components are strongly connected and the arborescence count does not depend on the root, and the nullity of L_c on $\text{supp } c$ is the number of those components [2, Lemma 3]. Deleting $r \notin \text{supp } c$ leaves the whole singular block L_c on $\text{supp } c$ intact, so the determinant is 0. $\square$

So a fixed root is legitimate on any part of the box on which it lies in the support. Such a part is produced, for free, by the symmetry of the problem.

Lemma 4 (ρ -symmetry). *$c \mapsto m - c$ is an involution of the set of circulations with $c + \rho c = m$, and $N_{\text{seq}}(m - c) = N_{\text{seq}}(c)$.*

Proof. $(m - c)(a) = c(\rho a)$, so $a \mapsto \rho a$, $v \mapsto \bar{v}$ carries G_{m-c} isomorphically to G_c with all arcs reversed, because $t(\rho a) = \bar{h(a)}$ and $h(\rho a) = \bar{t(a)}$. Hence $d_{m-c}^+(v) = d_c^-(\bar{v}) = d_c^+(\bar{v})$, so Loc is unchanged; $\prod_a c(a)!$ is unchanged because the multiset of values is permuted; and $t(G_{m-c})$ is the number of spanning *in*-trees of G_c , which for a balanced digraph equals the number of out-trees. $\square$

Corollary 5. *Let j_0 be a ρ -orbit with m_{j_0} odd. Then $f_{j_0} \neq 0$ on the box, the involution of Lemma 4 exchanges $\{f_{j_0} > 0\}$ with $\{f_{j_0} < 0\}$, and*

$$\#DS(S, k) = 2 \sum_{f: f_{j_0} > 0} N_{\text{seq}}\left(\frac{m+f}{2}\right).$$

On that half-box $c(a_{j_0}) \geq 1$, so $t(a_{j_0}) \in \text{supp } c$ for every point of it and the single vertex $r = t(a_{j_0})$ serves as root throughout.

At $k = 10$, 96 of the 111 orbits of $\phi X174$ have odd multiplicity, so the corollary applies; it halves the work and removes the root from the list of things the sweep has to decide.

4 What the sweep has to carry besides the fermions

Two more quantities in (1) are not arc-local.

Loc. $(d_c^+(v) - 1)!$ needs the whole out-degree of v at once. The partial out-degree is therefore carried as well — but only on the sub-frontier of the vertices whose out-arcs are not all decided, which at $k = 10$ is 18 wide against the 26 of the frontier itself, and is narrower still under an order chosen with it in mind. It is spent, and dropped, at the step at which the last out-arc of v is decided; if that out-degree is 0 the vertex is outside $\text{supp } c$ and the factor $(1 - \bar{\psi}_v \psi_v)$ of (3) is applied instead.

Integrality. $\prod_a c(a)!$ has no reason to divide anything along the way. It is removed from the sweep altogether: on a free orbit $c(a_j) + c(\rho a_j) = m_j$, so

$$\prod_a \frac{1}{c(a)!} = \frac{\prod_j \binom{m_j}{c(a_j)}}{\prod_j m_j! \cdot \prod_{\rho a=a} (m_a/2)!}, \quad (4)$$

whose numerator is a product of integer weights, one per orbit, and whose denominator is a constant of D_k . The sweep therefore runs over $\mathbb{Z}$ and divides once, at the end.

5 The sweep

The order of the orbits is Bio 12's greedy order, re-tuned: the elements the frontier now has to carry are one ψ per vertex over its whole window, weighted for the imbalance it travels with, and one $\bar{\psi}$ per vertex over its out-arc window, weighted for the partial out-degree. A short local search on top of a randomised greedy brings $\phi X174$ at $k = 10$ from 42 elements to 35.

The state of the sweep is

$$(\text{imbalance } b_v)_{v \text{ open}} \times (\text{partial out-degree})_{v \text{ out-open}} \times (\text{which } \bar{\psi}_v, \psi_v \text{ are spent}),$$

and its value is an integer. Each vertex is given a *slot* for the length of its window, by colouring the windows as intervals; slots are reused, so the layout is fixed once and for all and nothing has to be relabelled from step to step. The imbalances live in fixed-width fields of one machine word, biased so that an empty field reads zero, and the capacity bound of Bio 12 — $|b_v|$ at most the capacity still to come — is tested on all fields at once by two additions and two masks, which also forces a vertex leaving the frontier to balance. The out-arc window of v sits inside its full window, so $\bar{\psi}_v$ and ψ_v can share v 's slot; giving them the two bits $2s$ and $2s + 1$ of that slot makes the generator order interleaved, which by Remark 2 is what the constant of Proposition 1 requires. The signs are then local: multiplying on the right by a generator costs (-1) to the number of spent generators above it, and integrating one out when its window closes costs (-1)

to the number of spent generators below it, both read off the bit mask; the permutation between the order in which the generators leave and the canonical one is a single sign computed once.

Finally, the half-box may be cut into disjoint sub-boxes by pinning the values of a few early orbits, and the sums added. This changes nothing but the peak memory.

The sweep reproduces $\#DS = 10$ at $k = 12$ and 864 at $k = 11$ from the half-box, with one fixed root and one division at the end. It also agrees with the brute-force sum on twelve random covers. At $k = 11$ it peaks at 560 states.

6 Why the frontier stops short of $k = 10$

The frontier state of Section 5 is *linear*. At a fixed imbalance it is a vector in the exterior algebra of the open generators, and summing over histories adds vectors. The alphabet is small, but the vectors are not sparse. A boundary vertex v whose out-window is still open, and which already carries an arc to an open vertex u , contributes the factor $1 - \bar{\psi}_v\psi_v + \bar{\psi}_v\psi_u$. A single history with s such vertices therefore already has up to 3^s non-zero coordinates, before any two histories are added. At $k = 10$ the tuned order still has 35 open generators at its widest, and the weighted width estimate of the order search is $2^{86.8}$. The cancellations of the Berezin integral occur only at the end; along the way, the linear state pays for all the signed terms that will cancel.

The obvious alternative is the positive expansion of the same determinant, $t(G_c) = \sum_T \prod_{a \in T} c(a)$ over in-arborescences T of $\text{supp } c$ rooted at r . It needs no signs, and it puts *one* state on each history.

Proposition 6 (arborescence elimination). *Process the orbits in order. Call a vertex pending until its last out-arc has been decided, and let every pending vertex be the root of its own class. Carry, for each open vertex, its imbalance. For a pending vertex, also carry its candidate arcs so far as a multiset over the classes their heads belong to; its partial out-degree is their total. For a vertex that is no longer pending, carry the class it belongs to. When the last out-arc of a pending vertex $v \neq r$ has been decided, do one of two things. If its out-degree is 0, declare it outside $\text{supp } c$. Otherwise let it choose a candidate class K other than its own, with weight the multiplicity towards K times $(d_c^+(v) - 1)!$, and merge its class into K . The root never chooses; it contributes $(d_c^+(r) - 1)!$. Then the sum of the weights over all runs equals $\sum_c \prod_j \binom{m_j}{c(a_j)} t(G_c) \text{Loc}(G_c)$ over the half-box.*

Proof. A choice is an arc from v into a class other than v 's own. Following the choices from any vertex eventually reaches a pending vertex or r , so a cycle in the chosen arcs would have a last vertex to choose; that vertex would have found its head already in its own class. Conversely, a choice into a different class never closes a cycle. So the runs that survive correspond exactly to the in-arborescences of $\text{supp } c$ to r , each counted with $\prod_{a \in T} c(a)$. A component of the support that does not contain r has a last pending vertex whose candidates all lie in its own class, and so it contributes nothing. $\square$

It is exact. It reproduces 10 and 864, the random covers, and itself as a sum over sub-boxes. At $k = 10$ it agrees with the point-by-point sum of N_{seq} on five random sub-boxes of the half-box. But it merges too few histories. Those five sub-boxes hold between 12 and 166 points, with 30 to 36 of the 111 orbits pinned, and they already peak at between 55 890 and 623 000 states. The candidate multisets and class pointers are strictly finer than the connectivity partition with which Bio 12 counted the connected points, and that elimination already needed 9.7 million states. Neither frontier reaches the whole half-box on a 16 GB machine.

7 Point by point

What does reach it is the least clever method, made cheap at each point. The half-box has 1 240 843 950 points: half of Bio 12's count, by Corollary 5. The support of a point has about 126 arcs on about 110 vertices, so most vertices have out-degree 1, and the determinant is almost entirely forced.

Lemma 7 (forced arcs). *Let G be a balanced multidigraph, r a vertex, and $v \neq r$ a vertex all of whose out-arcs (there are $w \geq 1$ of them) end at one vertex $u \neq v$. Then $t_r(G) = w t_r(G/vu)$, where G/vu identifies v with u , deletes the w arcs $v \rightarrow u$ and redirects the arcs into v to u .*

Proof. Every in-arborescence to r contains exactly one out-arc of v , hence one of the w arcs to u . Contracting it is a bijection onto the in-arborescences of G/vu to r ; the other copies of $v \rightarrow u$, and the arcs from u to v , become loops, which no arborescence uses. $\square$

Applying the lemma until no vertex qualifies leaves a reduced Laplacian of small order. At each point, then:

1. the support is tested for connectivity by union-find, and discarded if it is not connected (Lemma 3);
2. $t(G_c)$ is computed by Lemma 7 followed by Gaussian elimination on what is left;
3. $\text{Loc}(G_c) / \prod_a c(a)!$ is read from tables.

All of this is done modulo the five largest primes below 2^{31} , so every product fits in a machine word. The points are visited by Bio 8's imbalance-bounded backtracking in Bio 12's greedy order, compiled with `numba`. The first 35 levels are expanded into 28 512 prefixes, which are shuffled and processed in parallel. The half-sum is recovered by the Chinese remainder theorem. This is exact: the half-sum is at most half of Bio 8's upper bound $1.368 \cdot 10^{36}$, far below the product of the five primes (about 2^{155}).

The run is checked in three ways, all exactly:

1. At $k = 12$ and $k = 11$ it returns 5 and 264 half-box points, 4 and 109 of them connected, and $\#DS = 10$ and 864; on twelve random covers it agrees with the brute force.
2. On $k = 10$ sub-boxes it agrees with the arborescence elimination of Proposition 6. That is a second, unrelated algorithm, and the sub-boxes are built from the same order, windows and root as a full run would use.
3. At $k = 10$ it counts exactly half of Bio 12's box points and exactly half of its connected points (Corollary 5; the involution preserves connectivity and has no fixed point on the box).

8 $\phi X174$ at $k = 10$

The root is vertex 20, the tail of orbit 33, which has multiplicity 1. The enumeration visits

$1\,240\,843\,950 = \frac{1}{2} \cdot 2\,481\,687\,900$ points, of which $454\,551\,605 = \frac{1}{2} \cdot 909\,103\,210$ have connected support,

k	rank Λ^-	box points	connected	#DS
12	4	10	8	10
11	12	528	218	864
10	48	2 481 687 900	909 103 210	31 925 753 246 212

Table 1: Bio 8’s Table 2, completed. The first two columns at $k = 10$ are Bio 12’s; the last is this note’s.

with no overflow of the work arrays. It takes about thirty-five minutes on sixteen threads. The sum of N_{seq} over the half-box, doubled, is

$$\#DS(\phi X174, 10) = 31\,925\,753\,246\,212 \approx 3.19 \cdot 10^{13}.$$

This lies inside Bio 8’s bounds $8\,610\,708\,632 \leq \#DS \leq 1.368 \cdot 10^{36}$: about 3 700 times the lower bound and twenty-three orders of magnitude below the upper one. Averaged over the 909 103 210 points with connected support, a point carries about 35 118 circular sequences. At $k = 11$ the average is $864/218 \approx 4$.

So at word length 10, $\phi X174$ is one of about $3.2 \cdot 10^{13}$ circular sequences with the same double-stranded 10-mer spectrum. The lattice has rank 48, but the count is not astronomical: the box has $2.5 \cdot 10^9$ points, and each carries a modest number of Eulerian circuits.

Ancillary files

`ds_fermion.py` is standalone: Python 3.10+ and the standard library. It rebuilds the double cover from its definition. It checks Proposition 1 against the reduced determinant point by point on $\phi X174$ at $k = 12$ and on twelve small random covers, and it checks Lemma 3, Lemma 4 and Corollary 5 on the enumerable boxes. It reproduces $\#DS = 10$ and 864 with the fermionic sweep and with the arborescence elimination, and it checks the latter, and Lemma 7, on $k = 10$ sub-boxes: seven checks in about ten minutes.

`ds_count.py` needs `numpy` and `numba`, and imports `ds_fermion.py`. It is the enumeration of Section 7 with its three checks, and it computes $\#DS(\phi X174, 10)$. The run takes about thirty-five minutes on sixteen threads and resumes from a checkpoint.

References

- [1] R. Chen, *Double-stranded reconstruction and the lattice of a reverse-complement double cover* (Bio 8 / XXI).
- [2] R. Chen, *The double-stranded lattice sum by frontier elimination* (Bio 12 / XXV).